

DEVANG KULSHRESHTHA

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Summary

Applied Scientist (AWS) with 5+ years in ML research and engineering, **10+ publications**, and **6 patents**. Focused on LLMs, Agentic AI, AI Safety, and Speech–Language Understanding. Proven record of advancing state-of-the-art systems, publishing research, and building solutions impacting millions.

Education

McGill University, *MSc in Computer Science* 2020–2022
Thesis: Neural Question Generation for Domain-Adaptive QA Systems | GPA: 4.0/4.0
Indian Institute of Technology (IIT) Varanasi, *B.Tech in Computer Science* 2014–2018
Top 10% of the class (Distinction) | GPA: 9.11/10

Work Experience

Amazon AWS AI (AgentCore, Bedrock, Lex) 2022–Present
Applied Scientist II

- Built AWS Bedrock Prompt Attack Detection, a guardrail LLM safety system outperforming Microsoft/Nvidia baselines by **35%** in adversarial robustness.
- Engineered the Policy Enforcement Engine for AWS AgentCore, achieving **90%** automated safety compliance across deployed agents.
- Implemented sequential editing–based continual training for speech models, reducing retraining compute cost by **45%** and enabling always-on adaptation in production.
- Led research on FAISS-retrieval based ASR personalization system for AWS Lex, scaling to **100K+** user catalogs and improving domain-specific transcription accuracy by **25%**.
- Supervised and mentored 3 research interns on projects in agent jailbreaking, LLM safety, and multi-agent social behavior modeling, resulting in submissions to top-tier AI conferences.

Amazon Alexa AI Summer 2021
Applied Science Intern

- Improved long-tail ASR recognition accuracy by **12%** via large language model rescoring. Results were integrated into Alexa’s multilingual ASR stack; received full-time return offer.

Korbit.AI (Intelligent Tutoring Systems) 2020–2022
Research Intern

- Designed and deployed transformer and retrieval-based models for automated question generation and adaptive feedback, powering large-scale ITS in production.

Amazon 2018–2020
Software Engineer

- Built scalable anomaly detection systems for invoice monitoring, reducing false positives by **30%**.
- Provided high-availability system support and mentored junior developers.

Selected Publications and Patents

- STAC: When Innocent Tools Form Dangerous Chains to Jailbreak LLM Agents **ICLR’26** (*submitted*) J.J. Li, J. He, C. Shang, D. Kulshreshtha, X. Xian, Y. Zhang, H. Su, S. Swamy, Y. Qi. [\[arXiv\]](#)
- Sequential Editing for Lifelong Training of Speech Recognition Models **INTERSPEECH 2024** D. Kulshreshtha, S. Dingliwal, B. Houston, N. Pappas, S. Ronanki. [\[Paper\]](#)
- Retrieve and Copy: Scaling ASR Personalization to Large Catalogs **EMNLP 2023** SM. Jayanthi, D. Kulshreshtha, S. Dingliwal, S. Ronanki, S. Bodapati. [\[Paper\]](#)
- Back-Training excels Self-Training at Unsupervised Domain Adaptation **EMNLP 2021** D. Kulshreshtha*, R. Belfer, I.V. Serban, S. Reddy. [\[Paper\]](#) [\[Presentation\]](#) [\[Code\]](#) [\[Poster\]](#)
- Removing bias from speech recognition models using internal language model estimates. [\[US Patent\]](#) N. Das, M.N. Sunkara, S.B. Bodapati, J. Cai, D. Kulshreshtha, J. Farris, N. Aldridge.
- Automated evaluation of evidence mapping models. [\[US Patent\]](#) D. Kulshreshtha, S. Dingliwal, S.B. Bodapati, K. Kirchhoff, S. Handa.
- Medical conversation summarization style intelligence. [\[US Patent\]](#) A. Elangovan, D. Kulshreshtha, K. Kirchhoff, L. Xu, S. Handa, S.B. Bodapati.